E-Commerce Sales & Customer Analytics

PROJECT DOCUMENTATION

An end-to-end data analytics project using Python, PostgreSQL, and Power BI

Project type	End-to-end data analytics
Dataset	7,478 cleaned rows × 17 columns
Tools	Python · Pandas · NumPy · Matplotlib · Seaborn · PostgreSQL · Power BI

Version 1 · Prepared as a project reference and interview-revision document

1. Project Overview

This project focuses on analyzing e-commerce sales and customer data to understand business performance. It involves cleaning and preparing the data, performing exploratory data analysis (EDA), analyzing business metrics using SQL, and creating an interactive Power BI dashboard.

The main purpose is to convert raw data into meaningful information that can help a business understand revenue, customer behavior, product performance, and order trends.

2. Project Objectives

- Clean and prepare raw sales data for analysis.
- Identify and handle missing values, duplicate records, and invalid data.
- Analyze sales revenue and order performance.
- Understand customer distribution and purchasing behavior.
- Compare revenue across product categories and cities.
- Analyze order distribution by payment mode and order status.
- Compare new and returning customers.
- Identify the top five products by revenue.
- Create an interactive dashboard to present key business metrics.

3. Dataset Description

The dataset contains order, customer, product, payment, and sales information. It initially contained approximately 7,500 records. After cleaning, the final dataset contained **7,478 rows and 17 columns**.

Column	Description
Order_ID	Order identifier
Order_Date	Date of the order
Customer_ID	Customer identifier
Gender / Age	Customer demographic fields
City	Customer's city
Category / Product	Product category and product name
Quantity	Number of units ordered
Unit_Price	Price per unit
Discount	Discount applied to the order
Payment_Mode	Payment method
Order_Status	Order status
Customer_Type	New or returning customer
Rating / Shipping_Days	Customer rating and shipping duration
Revenue	Calculated revenue for the order line

4. Tools and Their Purpose

Python: Used for data cleaning, transformation, exploratory analysis, and calculating business metrics.

Pandas and NumPy: Used to inspect data, handle missing values and duplicates, correct data types, and perform calculations.

Matplotlib and Seaborn: Used to create visualizations during exploratory data analysis.

PostgreSQL: Used to store the cleaned dataset and perform SQL-based analysis.

Power BI: Used to create an interactive dashboard with KPI cards, charts, and slicers.

5. Project Workflow

- **Data preparation:** Load the dataset and inspect its structure, columns, data types, and quality.
- **Data cleaning:** Handle missing values, remove duplicate rows, standardize inconsistent text, and remove invalid quantity or price records.
- **Exploratory data analysis:** Examine distributions, categories, revenue patterns, and customer-related trends.
- **Database integration:** Load the cleaned data into PostgreSQL for SQL analysis.
- **Business analysis:** Use aggregations and business metrics to analyze sales and customer performance.
- **Dashboard development:** Build Power BI visuals and interactive slicers.
- **Final review:** Check accuracy, readability, and usability.

6. Data Cleaning

Missing values: Missing values were inspected and handled according to the column's meaning and suitability for imputation.

Duplicate records: Duplicate rows were removed to prevent repeated records from distorting the analysis.

Invalid quantities: Records with zero or negative quantities were removed.

Invalid prices: Records with zero or negative unit prices were removed.

Inconsistent text: Category and other text values were standardized to avoid splitting the same category across different spellings or formats.

Revenue calculation: Revenue was calculated using quantity, unit price, and discount.

Revenue formula

$$\text{Revenue} = \text{Quantity} \times \text{Unit Price} \times (1 - \text{Discount})$$

This formula assumes the discount is stored as a decimal fraction—for example, 0.10 represents a 10% discount.

7. Power BI Dashboard

KPI cards

- Total Revenue
- Total Orders
- Average Order Value
- Total Customers

Analytical visuals

- **Monthly Revenue Trend:** tracks revenue over time.

- **Revenue by Category:** compares category performance using a treemap.
- **Revenue by City:** compares revenue across cities using a bar chart.
- **Orders by Payment Mode:** compares orders across payment methods.
- **Orders by Status:** shows order-status distribution using a donut chart.
- **New vs Returning Customers:** compares customer types using a 100% stacked column chart.
- **Top 5 Products by Revenue:** identifies the five highest-revenue products using a bar chart.

Interactive slicers: Slicers allow users to filter the dashboard and inspect selected portions of the data. Their interactions were tested to confirm that visuals update when filters change.

8. Expected Business Value

- Monitor overall revenue and order performance.
- Compare sales across categories and cities.
- Identify products generating the most revenue.
- Understand payment-method and order-status distributions.
- Compare new and returning customer groups.
- Explore trends by applying interactive filters.

9. Project Conclusion

This project demonstrates an end-to-end data analytics workflow, from data preparation and cleaning to database analysis and dashboard development. It combines Python, PostgreSQL, and Power BI to organize data, calculate key metrics, and present business information in an accessible format.

The project provides practical experience in data quality management, exploratory data analysis, KPI development, SQL-based analysis, and data visualization.

Important: Add specific business findings and recommendations only after validating the final analysis results. Check dashboard metrics against the underlying dataset before presenting the project in an interview.